

Design and Physical Sign-off of a RISC-V SoC (RV32IM) with a DOOM Port on SkyWater 130 nm via OpenLane 2

An RTL-to-GDS study of an open RISC-V system-on-chip

RISCV Social Project · Domains: RTL, FPGA (ECP5), ASIC (SKY130)

Abstract—

This paper documents the design and end-to-end physical sign-off of **riscv_doom_soc**, an open RISC-V RV32IM system-on-chip with an on-chip timer, UART, QSPI flash, SPI-TFT display controller, and a DOOM game port. The project spans RTL development, behavioral verification with Icarus Verilog, software bring-up with the GNU RISC-V toolchain, ECP5 FPGA tooling, and culminates in a full **RTL to GDS-II** flow through **OpenLane 2** on the **SkyWater 130 nm (sky130B)** library, verified with Magic, KLayout, and Netgen. The completed run closes with **zero DRC violations, zero LVS mismatches, and timing met** at 20 ns on the nominal corner, producing a merged GDS-II. We report synthesis area, P&R statistics, CTS metrics, power, and timing, and discuss OpenLane 2 behavior on a register/SRAM-heavy design.

I. Introduction

Open and silicon-verifiable RISC-V cores are a cornerstone of modern IC design. The SkyWater 130 nm open PDK and the OpenLane autonomous RTL-to-GDS flows have lowered the barrier to tape-out-eligible layouts from Verilog. This work integrates an in-house RV32IM core with a set of peripherals and a DOOM-compatible platform, and demonstrates that the same RTL is capable of production-style physical-design closure.

Our contribution is (1) the architecture and bring-up of a complete RISC-V SoC, and (2) a careful documentation of the OpenLane 2 flow, including the interaction of synthesized memories with timing repair. We present measured GDS-II closure results and the lessons learned when a flop-based SRAM drives the post-CTS timing-resizer workload.

II. Architecture

The SoC is a Harvard RV32IM design: a boot-ROM, a 32 KB on-die memory (synthesized to flops in this flow), a QSPI flash controller for execute-in-place, a UART, a

timer with interrupt, and an SPI-TFT (ILI9341) controller. Table I lists the physical memory map.

Table I: Memory map

Address	Region
0x0000_0000	BootROM (4 KB logic)
0x0001_0000	On-chip SRAM 32 KB (OpenRAM macro target)
0x1000_0000	QSPI-flash XIP window (8 MB)
0x2000_0000	QSPI-flash XIP alt
0x4000_0000	QSPI controller regs
0x4001_0000	SPI-TFT (ILI9341) + pixel DMA
0x4002_0000	UART
0x4003_0000	Timer (mtime/mtimecmp) + IRQ

RTL lives under **rtl/rv32** (ALU, decoder, CSR, immgen, M-extension mul/div, register file, core), **rtl/soc** (boot-ROM, SRAM wrap, glue), and **rtl/periph** (QSPI, SPI-TFT, UART, timer). Self-checking Icarus Verilog testbenches verify the core, the M-extension, the full SoC, and the QSPI engine. The software stack — CRT0, linker script, platform HAL — is built with the GNU RISC-V toolchain and drives a DOOM-compatible demo.

III. Physical-design flow

The physical flow uses **OpenLane 2 (2.3.10)** in a Docker container on the SkyWater 130 nm open PDK (sky130B, volare 0fe599b), with sky130_fd_sc_hd as the standard-cell library. The Classic stage list runs: Verilator lint, Yosys+ABC synthesis, OpenROAD floorplan / placement / CTS / routing, timing resizing, and finally Magic & DRC, Netgen LVS, and GDS-II stream-out.

Table II: Synthesis (full SoC, Yosys)

Metric	Value
Design	riscv_doom_soc
Synthesized cells	68,690
Chip area	916,336 um2 (~0.92 mm2)
Sequential fraction	43.05 % (synthesized SRAM / regfile)

IV. Sign-off results (tractable probe)

To demonstrate complete closure -- through post-CTS resizing, routing, DRC, LVS, and GDS-II -- a minimal-RAM variant of the SoC was taken to completion. Table III summarizes the measured final metrics; the detailed router converged its DRC errors to zero across 13 global-routing iterations.

Table III: Final GDS-II sign-off metrics

Metric	Value
Standard cells (instances)	18,267

Die size (bbox)	522.5 x 533.2 um (0.28 mm ²)
Core utilization	60.9 %
Total power	8.29 mW
Clock period (target)	20 ns (50 MHz)
Setup slack (nom TT)	+8.84 ns; WNS=0, TNS=0
Hold slack	+0.29 ns; WNS=0, TNS=0
Setup / Hold violations	0 / 0
Wirelength (routed)	698,942 um
Vias	124,148
DRC errors (Magic)	0
DRC errors (KLayout)	0
LVS (device / net diff)	0 / 0 (pass)
Antenna violations	71 nets -> 452 diodes

V. Discussion

The dominant runtime is the post-CTS resizer on the synthesized memory. Because the 32 KB SRAM and register file elaborate to 18,000+ flops, the resizer must repair a very large endpoint set (17,809 setup endpoints in the full design). An OpenRAM hard-memory macro in place of flop SRAM is the recommended next step. On the slow (SS) corner setup closes at -2.13 ns; relaxing the clock to ~25 ns closes the slow-corner path while nominal/fast corners have ample margin.

VI. Conclusion

We presented an open RISC-V RV32IM SoC with a DOOM port and demonstrated its complete physical-design closure to GDS-II through OpenLane 2 on SkyWater 130 nm. The layout is DRC-clean, LVS-clean, and timing-clean on the nominal corner. The GDS-II, DEF, and metrics are archived.

References

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